

Welcome to Systematic Design

SEAtS | 544403



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Lectures, Assignment Workshops



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Lectures, Formal Methods workshop

Purpose of this Module

The goal of this module is to give students an overview of the theory and methods for designing and delivering large multi-component systems.

There will be a particular emphasis on systems which combine requirements and functionality from learning, software, and hardware, and the ways in which the boundaries between them can be identified and managed.

This module provides preparatory tools and techniques for the group project module in semester 2.

Motivation



Syllabus

Analysing systems

- Functional and non-functional requirements
- Requirements gathering
- System architecture

Designing systems

- Decomposing into blocks
- Identifying interfaces
- Designing APIs and protocols
- Integrating libraries and IP
- Formal specification

Testing systems

- Build systems
- Continuous integration (both hw and sw)
- Testing hardware+software+network systems
- Formal verification

Managing development

- Development methodologies
- Project management
- Advanced source control
- (e.g. review, bots, ...)

Course Structure

Three 1-hour lectures per week

- Tuesday 09:00 (35/1005)
- Tuesday 17:00 (35/1005)
- Thursday 14:00 (02/1039)

One 3-hour Workshop per week

- Wednesday 09:00–12:00 (02/1039)

Coursework and Exams

Coursework

- (Tentative) Set in Week 4
- You can start early, but some material up to Week 8 will be required.
- Due in Week 11 (before the X-mas break)
- Feedback in the new year

Final exam

- Semester 1 Exam period (in the New Year)
- 12-hour Computer-Aided Assessment (CAA)

Course Management

Moodle

(Required login with your University ID)

<https://moodle.ecs.soton.ac.uk/course/view.php?id=1022>

▶ Annoucements will be sent via Moodle.

Timetables

 **Your timetable shows all of the sessions/workshops allocated to your modules**

 **Modules do not always use all of those sessions!**

- For example, we have a Reading Week

 **So check the notes pages of each course for more info and adjust your timetable accordingly!**

Reminder AICE2005 Tentative Timetable

How to Get Help

- Refrain from emailing (get lost in the avalanche)
- Don't drop by the office without schedule (often busy)
- Do come down to ask questions at the end of the lecture
- Do ask questions in the workshop
- Do ask each other!
- Be careful of academic responsibility and conduct.

Your Special Circumstances

If you have a special circumstance that might affect your assessments (e.g., coursework, exams):

- Don't panic
- Don't ask us (the module team) for extensions, etc.
- Do talk to your personal academic tutor
- Do apply to the Special Circumstance Board (SCB).
- You need evidence to support your claim, e.g., doctor letter
- Contact the Student Hub to fill out the relevant forms
- To extend the deadline (if the deadline has not yet passed)
- To waive the late penalty (if the deadline has already passed).
- To resit an exam

Our Special Circumstances

If we have a special circumstance that affects the delivery of a lecture:

- Don't panic
- If feasible, we (the module team) will arrange to cover lecture
- Else, we will rearrange the timetable if it makes sense
 - (We have a couple of free back-up slots)
- Otherwise, we will provide a lecture recording for the missing lecture.

Thank you

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